7.1 p. 399 #1, 5, 7, 9, 13, 17, 23, 25

1. $\begin{bmatrix} 3 & 5 \\ 5 & -7 \end{bmatrix}$ yes

5. $\begin{bmatrix} -6 & 2 & 0 \\ 0 & -6 & 2 \\ 0 & 0 & -6 \end{bmatrix}$ no

7. $\begin{bmatrix} .6 & .8 \\ .8 & -.6 \end{bmatrix} = A$ $\quad .6(.8) + .8(-.6) = 0$ ✓orthogonal

$\sqrt{\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2} = \sqrt{\frac{9}{25} + \frac{16}{25}} = 1$ ✓square ✓orthonormal

$A^{-1} = A^T = \begin{bmatrix} .6 & .8 \\ .8 & -.6 \end{bmatrix}$

9. $\begin{bmatrix} -5 & 2 \\ 2 & 5 \end{bmatrix} = A$ $\quad -5(2) + 2(5) = 0$ ✓orthogonal

✓square

$\sqrt{5^2 + 2^2} = \sqrt{29} \neq 1$ xnormal

∴ not orthogonal matrix

13. $\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$ $\quad \begin{vmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{vmatrix} = (3-\lambda)^2 - 1 = 0$

$\lambda^2 - 6\lambda + 8 = 0$

$(\lambda - 4)(\lambda - 2) = 0$

$\lambda = 4, 2$

$\lambda = 4$: $\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$ $v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ $\|v_1\| = \sqrt{2}$

$\lambda = 2$: $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ $v_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ $\|v_2\| = \sqrt{2}$

$P = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$ $\quad D = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}$

17. $\lambda = 5, 2, -2$

$A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & 1 \\ 3 & 1 & 1 \end{bmatrix}$ $\quad \lambda = 5$: $\begin{bmatrix} -4 & 1 & 3 \\ 1 & -2 & 1 \\ 3 & 1 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$ $\quad v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ $\|v_1\| = \sqrt{3}$

$\lambda = 2$: $\begin{bmatrix} -1 & 1 & 3 \\ 1 & 1 & 1 \\ 3 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$ $\quad v_2 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$ $\quad \|v_2\| = \sqrt{6}$

$\lambda = -2$: $\begin{bmatrix} 3 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $\quad v_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ $\quad \|v_3\| = \sqrt{2}$

$P = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{6} & -1/\sqrt{2} \\ 1/\sqrt{3} & -2/\sqrt{6} & 0 \\ 1/\sqrt{3} & 1/\sqrt{6} & 1/\sqrt{2} \end{bmatrix}$ $\quad D = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$

23. $A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{bmatrix}$ $\quad v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$? $\quad Av_1 = \begin{bmatrix} 5 \\ 5 \\ 5 \end{bmatrix} = 5v$ $\quad \therefore \lambda_1 = 5$

$v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ $\quad \|v_1\| = \sqrt{3}$

$\lambda = 2$: $x_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ $\quad x_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ $\quad \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

non trivial solutions so $\lambda_2 = 2$ w/ mult 2 (2 free $\therefore$ dim eigen space = 2)

$v_2 = x_2 = (-1, 1, 0)$ $\quad \|v_2\| = \sqrt{2}$

$x_3 - \frac{x_3 \cdot v_2}{v_2 \cdot v_2} v_2 = x_3 - \frac{1}{2} v_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} -1/2 \\ 1/2 \\ 0 \end{bmatrix} = \begin{bmatrix} -1/2 \\ -1/2 \\ 1 \end{bmatrix}$

$v_3 = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$ $\quad \|v_3\| = \sqrt{6}$

$P = \begin{bmatrix} 1/\sqrt{3} & -1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & 0 & -2/\sqrt{6} \end{bmatrix}$ $\quad D = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

25.

a. True (Thm 2)

b. True (symmetric, u, v eigenvectors, must be orthogonal) → Thm 1, u and v nonzero

c. False (n eigenvalues including multiplicities)

d. False (only if V is an orthonormal eigenvector)

7.2 p.406 #1,5,7,9,13

1. $A=\begin{bmatrix} 5 & 1/3 \\ 1/3 & 1 \end{bmatrix}$

a. $5x_1^2 + x_2^2 + \frac{2}{3}x_1x_2$

b. $5(6)^2 + 1^2 + \frac{2}{3}(6)(1) = 5(6^2)+5 = 185$

c. $5(1)^2 + 3^2 + \frac{2}{3}(1)(3) = 5+9+2 = 16$

5.

a) $8x_1^2+7x_2^2-3x_3^2-6x_1x_2+4x_1x_3-2x_2x_3$ $\begin{bmatrix} 8 & -3 & 2 \\ -3 & 7 & -1 \\ 2 & -1 & -3 \end{bmatrix}$

b) $4x_1x_2+6x_1x_3-8x_2x_3$ $\begin{bmatrix} 0 & 2 & 3 \\ 2 & 0 & -4 \\ 3 & -4 & 0 \end{bmatrix}$

7. $x_1^2+10x_1x_2+x_2^2$

$A=\begin{bmatrix} 1 & 5 \\ 5 & 1 \end{bmatrix}$ $\begin{vmatrix} 1-\lambda & 5 \\ 5 & 1-\lambda \end{vmatrix} = (1-\lambda)^2-25=0$ $\lambda=6,-4$

$\lambda^2-2\lambda-24=0$

$(\lambda-6)(\lambda+4)=0$

$\lambda=6$: $\begin{bmatrix} -5 & 5 \\ 5 & -5 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$ $v_1=\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ $\|v_1\|=\sqrt{2}$

$\lambda=-4$: $\begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ $v_2=\begin{bmatrix} -1 \\ 1 \end{bmatrix}$ $\|v_2\|=\sqrt{2}$

$P=\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$ $D=\begin{bmatrix} 6 & 0 \\ 0 & -4 \end{bmatrix}$ $6y_1^2-4y_2^2$

→

* multiple answers possible depending on eigenvectors (i.e. book has y_1 and y_2 switched)

9. $3x_1^2 - 4x_1x_2 + 6x_2^2$

$A = \begin{bmatrix} 3 & -2 \\ -2 & 6 \end{bmatrix}$ $\quad \begin{vmatrix} 3-\lambda & -2 \\ -2 & 6-\lambda \end{vmatrix} = (3-\lambda)(6-\lambda) - 4 = 0$

$\lambda^2 - 9\lambda + 14 = 0$

$(\lambda - 7)(\lambda - 2) = 0$

positive definite ← $\quad \lambda = 2, 7$

$\lambda = 2$: $\begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix}$ $\quad v_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ $\quad \|v_1\| = \sqrt{5}$

$\lambda = 7$: $\begin{bmatrix} -4 & -2 \\ -2 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1/2 \\ 0 & 0 \end{bmatrix}$ $\quad v_2 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ $\quad \|v_2\| = \sqrt{5}$

$P = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}$ $\quad$ w/ $\vec{x} = P\vec{y}$ $\quad D = \begin{bmatrix} 2 & 0 \\ 0 & 7 \end{bmatrix}$ $\quad$ so $2y_1^2 + 7y_2^2$

13. $x_1^2 - 6x_1x_2 + 9x_2^2$

$A = \begin{bmatrix} 1 & -3 \\ -3 & 9 \end{bmatrix}$ $\quad \begin{vmatrix} 1-\lambda & -3 \\ -3 & 9-\lambda \end{vmatrix} = (1-\lambda)(9-\lambda) - 9 = 0$

$\lambda^2 - 10\lambda = 0$

$\lambda(\lambda - 10) = 0$

positive semidefinite ← $\quad \lambda = 0, 10$

$\lambda = 0$: $\begin{bmatrix} 1 & -3 \\ -3 & 9 \end{bmatrix} \sim \begin{bmatrix} 1 & -3 \\ 0 & 0 \end{bmatrix}$ $\quad v_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ $\quad \|v_1\| = \sqrt{10}$

$\lambda = 10$: $\begin{bmatrix} -9 & -3 \\ -3 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1/3 \\ 0 & 0 \end{bmatrix}$ $\quad v_2 = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$ $\quad \|v_2\| = \sqrt{10}$

$P = \frac{1}{\sqrt{10}} \begin{bmatrix} 3 & -1 \\ 1 & 3 \end{bmatrix}$ $\quad$ w/ $\vec{x} = P\vec{y}$ $\quad D = \begin{bmatrix} 0 & 0 \\ 0 & 10 \end{bmatrix}$ $\quad$ so $10y_2^2$